

CIS 613: Testing Ethics

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Professional vs General or Personal Ethics

- Two ways professional ethics differ from general ethics:
 - Professional is an **expert** in a field and therefore has special responsibilities.
 - Products of many professionals profoundly **affect large number of people**.

A professional can cause great harm through dishonesty, carelessness, or incompetence.

Adapted from Prof. Nandigam

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SE Code of Ethics and Professional Practices

- Developed by IEEE-CS and ACM Joint Task Force
- Eight principles (with several clauses for each principle)
 - Public
 - Client and Employer
 - Product
 - Judgement
 - Management
 - Profession
 - Colleagues
 - Self
- The SE Code should not be viewed as a simple ethical algorithm for generating ethical decisions.
- Software engineers should strive to act in a manner that is consistent with the spirit of the SE code.

Adapted from Prof. Nandigam

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SE Code of Ethics and Professional Practices

1. **Public** – Software Engineers shall act consistently with the public interest
2. **Client and Employer** – Software Engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest
3. **Product** – Software Engineers shall ensure their products and related modifications meet the highest professional standards possible.
4. **Judgement** – Software Engineers shall maintain integrity and independence in their professional judgement.

Adapted from Prof. Nandigam

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SE Code of Ethics and Professional Practices

5. **Management** – Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6. **Profession** – Software Engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7. **Colleagues** – Software Engineers shall be fair to and supportive of their colleagues.
8. **Self** – Software Engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Adapted from Prof. Nandigam

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ACM Code of Ethics

Does the ACM Code of Ethics *actually* encourage people to act ethically?

The code of ethics has been shown* to have no impact.

**Does ACM's code of ethics change ethical decision making in software development?, FSE 2018*

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Ethical Decision Making?

- In a study performed with:
 - 63 Software Engineering Students, and
 - 105 Professional Software Developers.
- Participants were asked to analyze 11 short ethical questions focusing on:
 - A control group, and
 - A group specifically requested to consider the ACM code of ethics

**There was no
statistical/observed effect in
the additional instructions!**

Should it have an impact?

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Ethical Decision Making

- Ethics is the study of what it means to **“do the right thing”**
- When faced with difficult ethical situations, it helps to restate an ethical issue, question or dilemma in the form:
 - **Is it right to ...?”**

Adapted from Prof. Nandigam

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Other Ethical Tests

- **Harm/Beneficence Test**
 - “Does it do less harm and more good than alternatives?”
 - e.g., Institutional Review Board
- **Publicity Test**
 - “Would I want this choice published in a newspaper?”
 - Presentation of self to others and self (consistency)
- **Reversibility Test**
 - “Would I think it was a good choice if I were affected by it?”
 - Empathy (also a good skill for design!)
- **Code of Ethics Test**
 - “What do the rules of my profession say?”

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Why Ethics?

- “Good ethical behavior usually **leads to good consequences**, both for ourselves and for society at large.”
- “Scientists and engineers make decisions **crucial to society at large**, and therefore shoulder an enormous burden of public trust.”
- “**Happiness** comes from reasoning through a complex moral puzzle, choosing a good course of action, and following through.”

Source: “Fundamentals of Ethics for Scientists and Engineers” by Seebauer & Barry

Adapted from Prof. Nandigam

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Ethical Decision Making for Testing

- Should **testing ethics** be the same as **software engineering ethics**?
- Would ethics ever require us to change the **amount of testing** we perform in order to ensure we had tested *ethically*?
- Would **cultural issues** ever change how we might think to act ethically?